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Reconciliation Checkpoints as SOC Hunt-Cycle Doctrine

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Abstract

The Skipjack Protocol's reconciliation checkpoint mechanism (Principle 4 of the HGC³AE² parent framework) is structurally **isomorphic** to the threat-hunt cycle that mature Security Operations Centers (SOCs) already practice. The isomorphism is not coincidental — both protocols address the same operational concern (continuous re-alignment of operator intent against an evolving operational state) in different domains (agentic system governance vs. defensive threat hunting). This paper develops the mapping and produces an **integration doctrine for AI-assisted defensive operations** that preserves human analyst authority over the hypothesis, the triage decision, and the escalation call, while allowing agentic tooling to execute within each phase.

The contribution is operationally concrete: mapping Skipjack checkpoints onto hunt-phase transitions (hypothesis → hunt, hunt → triage, triage → escalation) yields four named gates that translate directly into SOAR-compatible runbook policy. The reconciliation discipline scales human analyst capacity without surrendering analyst authority, addressing the automation-bias risk that emerges when analysts over-trust AI-triage outputs.

Section 2 establishes the SOC threat-hunt cycle and the Skipjack reconciliation principle. Section 3 develops the mapping with four named checkpoint gates (CP-1 Triage Gate, CP-2 Escalation Gate, CP-3 Session Resumption Gate, CP-4 Scope Authority Gate). Section 4 specifies governance dimensions: checkpoint cadence, escalation threshold governance, non-repudiation as a governance control. Section 5 supplies a runbook policy block (proposed) that SOAR platforms can adopt directly. Section 6 closes with limitations and the path to industry standards integration.

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ABSTRACT

The Skipjack Protocol's reconciliation checkpoint mechanism (Principle 4 of the HGC³AE² parent framework)¹ is structurally **isomorphic** to the threat-hunt cycle that mature Security Operations Centers (SOCs) already practice. The isomorphism is not coincidental — both protocols address the same operational concern (continuous re-alignment of operator intent against an evolving operational state) in different domains (agentic system governance vs. defensive threat hunting). This paper develops the mapping and produces an **integration doctrine for AI-assisted defensive operations** that preserves human analyst authority over the hypothesis, the triage decision, and the escalation call, while allowing agentic tooling to execute within each phase.

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The central claim is that **AI-assisted SOC operations need a reconciliation doctrine, not merely a triage automation layer**: triage automation without reconciliation surfaces produces analyst over-trust at exactly the moment when alignment-against-original-hypothesis matters most. The Skipjack checkpoint model supplies the missing surface.

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1. INTRODUCTION

1.1 THE AI-ASSISTED SOC

Modern Security Operations Centers increasingly deploy AI tooling for triage, hunting, and incident response. The tooling reduces analyst load on routine alert handling, surfaces hypothesis candidates from telemetry, and accelerates evidence assembly. The deployment patterns vary by maturity and tool selection; what is common across mature SOCs is that

1 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

2 Linda J. Skitka, Kathleen L. Mosier, and Mark Burdick, "Does Automation Bias Decision-Making?" *International Journal of Human-Computer Studies* 51, no. 5 (1999): 991-1006.



analyst authority over the hunt cycle is preserved as a governance requirement, regardless of how much tooling automation is present.

The tension this paper addresses is between automation-driven throughput and analyst-driven authority. Tooling that fires on alerts and pre-resolves triage decisions risks producing analyst over-trust:³⁴ the analyst accepts the tool's triage conclusion without re-aligning the conclusion against the original hunt hypothesis, and the original hypothesis (which is what the human analyst owned) gets quietly displaced by the tool's pattern-match.

1.2 THE CHECKPOINT PROBLEM

The structural problem is that **the hunt cycle has phase transitions where re-alignment matters**, and automated tooling typically does not pause at those transitions in a way that surfaces re-alignment as an analyst decision. The transitions:

Hypothesis → Hunt: the analyst commits to a hunt hypothesis; the tooling executes searches against that hypothesis. The transition is where hypothesis-vs-execution alignment is set.

Hunt → Triage: the tooling surfaces candidates; the analyst classifies them against the hypothesis. The transition is where the tool's pattern-matching is re-aligned against the human-owned hypothesis.

Triage → Escalation: the analyst decides which classified candidates warrant incident-response escalation. The transition is where the analyst's authority over operational consequence is exercised.

Each transition is a checkpoint that requires analyst-authority engagement. The Skipjack reconciliation principle⁵ is, structurally, the same discipline applied to general agentic operation: the system pauses at consequential transitions and re-seeks alignment with operator intent before proceeding.

1.3 CONTRIBUTION

This paper:

Develops the Skipjack ↔ hunt-cycle isomorphism with specific mappings between reconciliation principles and hunt-phase transitions.

Specifies four named SOC checkpoints (CP-1 through CP-4) that translate Skipjack discipline into SOC operational vocabulary.

Provides governance dimensions (cadence, threshold, non-repudiation) that production SOCs can adopt.

Supplies a SOAR-compatible runbook policy block that platforms can ingest as an executable governance artifact.

3 Linda J. Skitka, Kathleen L. Mosier, and Mark Burdick, "Does Automation Bias Decision-Making?" *International Journal of Human-Computer Studies* 51, no. 5 (1999): 991-1006.

4 Mary L. Cummings, "Automation Bias in Intelligent Time Critical Decision Support Systems," *Proceedings of the AIAA First Intelligent Systems Technical Conference*, 2004.

5 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.



2. BACKGROUND

2.1 THE SKIPJACK PROTOCOL — PRINCIPLE 4

The HGC³AE² parent framework specifies four Skipjack principles; Principle 4 — *iterative feedback / continuous reconciliation* — addresses the structural requirement that agentic execution must continuously re-align with operator intent rather than execute against an aging snapshot of intent.⁶ The reconciliation is operationalized through checkpoints: pause points at which the system surfaces its current understanding of intent and the operator either confirms, adjusts, or escalates.

2.2 THE SOC THREAT-HUNT CYCLE

Mature SOCs practice threat hunting as a structured cycle:⁷⁸

Hypothesis formation. Analyst-owned. A hypothesis is a structured claim about adversarial activity that telemetry could either support or rule out.

Hunt execution. Tooling-supported. Telemetry queries, behavioral analytics, indicator searches. The execution operates *under* the hypothesis.

Triage. Analyst-owned, tooling-supported. Surface candidates against the hypothesis; classify as true positive, false positive, or under-investigation.

Escalation. Analyst-owned. True positives become incident-response engagements.

Closure / learning. The cycle closes; lessons feed back into hypothesis formation for the next cycle.

The cycle is iterative; mature SOCs run multiple concurrent hunts at different phases.

2.3 AUTOMATION BIAS IN THE SOC

Automation bias⁹ is the human tendency to over-trust automated outputs in high-information-load environments. SOCs are precisely such environments — alert volumes vastly exceed manual analyst capacity, so analysts develop heuristics that favor automated classifications. The heuristic is operationally rational; the security consequence is that when the automation is wrong, the analyst's authority-over-outcome is silently transferred to the tooling.

The Skipjack reconciliation discipline addresses this by structurally requiring re-alignment at phase boundaries: the tool produces output, but the analyst's commit to next-phase action requires explicit re-engagement with the hypothesis.

6 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

7 Sqrri Data, *The Hunter's Handbook: Endgame's Guide to Adversary Hunting* (2017). (Canonical practitioner reference for the hunt cycle.)

8 David J. Bianco, "The Pyramid of Pain" (2013), <https://detect-respond.blogspot.com/2013/03/the-pyramid-of-pain.html>. (Cited for the analyst-authority taxonomy at different hunt-phase abstraction levels.)

9 Mary L. Cummings, "Automation Bias in Intelligent Time Critical Decision Support Systems," *Proceedings of the AIAA First Intelligent Systems Technical Conference*, 2004.



3. THE MAPPING: SKIPJACK CHECKPOINTS ONTO THE HUNT CYCLE

3.1 STRUCTURAL ALIGNMENT

The Skipjack reconciliation principle and the SOC hunt cycle share a structural commitment: **human authority over the analytical chain**, with automation operating within the chain rather than over it. The mapping below names four checkpoint gates corresponding to phase transitions in the hunt cycle.

3.2 CP-1: TRIAGE GATE (HUNT → TRIAGE)

Transition. Hunt execution completes; candidate set is surfaced; analyst commits to triage classifications.

Reconciliation requirement. Before classifications are committed, the analyst re-engages with the hunt hypothesis to confirm that each classification aligns with the hypothesis. Tool-suggested classifications are surfaced as recommendations; the analyst's commit is the authoritative classification.

Skipjack mapping. Principle 4 (continuous reconciliation) at the operator-intent-vs-tool-output boundary.

Operator authority preserved. The analyst owns each classification. The tool's recommendation is an input; the commit is the analyst's.

3.3 CP-2: ESCALATION GATE (TRIAGE → ESCALATION / IR HANDOFF)

Transition. Triage produces a classified candidate; analyst decides whether the candidate warrants incident-response escalation.

Reconciliation requirement. Escalation decisions re-engage the hypothesis AND the consequence chain: does this candidate, classified as true positive, justify the operational consequences of IR engagement (resource cost, business impact, response timeline)?

Skipjack mapping. Principle 4 + Principle 3 (explicit uncertainty signaling). The analyst signals confidence in the classification AND in the consequence chain.

Operator authority preserved. Escalation is the analyst's commit. The tool may produce recommendations and route candidate metadata; the operational decision remains the analyst's.

3.4 CP-3: SESSION RESUMPTION GATE (ANY RESUME)

Transition. A hunt is paused — shift change, analyst rotation, environment switch — and resumed. The resuming analyst inherits state from the prior analyst (or from the system's saved state).

Reconciliation requirement. Before resuming, the resuming analyst re-engages the hypothesis: is the hypothesis still valid given the elapsed time? Have conditions changed that should update the hypothesis? Is the prior analyst's state-of-work congruent with the resuming analyst's understanding?



Skipjack mapping. Principle 4 across persona-handoff events. This is the cyb-10 chain-of-custody discipline¹⁰ applied at SOC scale.

Operator authority preserved. The resuming analyst takes explicit authority. The session state is consult-able but not authoritative; the resuming analyst's re-engagement is the authoritative state.

3.5 CP-4: SCOPE AUTHORITY GATE (SCOPE OR STAKEHOLDER CHANGE)

Transition. Mid-hunt, the scope changes (new business unit involved, new asset class implicated, new stakeholder requires inclusion) or stakeholder authority shifts (analyst-on-call rotation, supervisor change).

Reconciliation requirement. Scope-change events re-engage the hypothesis with the new scope: does the hypothesis remain meaningful at the new scope? Does the new stakeholder authority alter the hypothesis or the response options?

Skipjack mapping. Principle 4 + Principle 1 (context integrity verification). The hypothesis's context has shifted; the operating model must verify integrity against the new context.

Operator authority preserved. The new stakeholder takes explicit authority at the scope change. Pre-existing tool actions are consult-able; the new authority's commit is what governs continuation.

4. GOVERNANCE DIMENSIONS: OPERATIONALIZING THE DOCTRINE

4.1 CHECKPOINT CADENCE

Checkpoints fire at phase transitions, not on a clock. The cadence is *event-driven* (hunt completion, triage commit, resumption event, scope change) with operator-pinned thresholds for each event class. Production SOCs configure thresholds based on operational risk tolerance and analyst capacity.

4.2 ESCALATION THRESHOLD GOVERNANCE

The CP-2 escalation gate has operator-pinned thresholds: which candidate classifications warrant escalation, what evidence quality is required, what response-team availability is assumed. The thresholds are versioned and audit-able; threshold changes are governance events with their own audit trail.

4.3 NON-REPUDIATION AS A GOVERNANCE CONTROL

Each checkpoint produces a *non-repudiable record*: the checkpoint event, the analyst's decision, the supporting evidence, the policy version active at decision time. The records compose into the cyb-10 chain-of-custody ledger¹¹ for the hunt cycle as a whole. The

¹⁰ Forthcoming CYB-10 *Human Authority Boundary as Chain-of-Custody for Agentic Outputs* (in concurrent drafting; this dispatch).

¹¹ Forthcoming CYB-10 *Human Authority Boundary as Chain-of-Custody for Agentic Outputs* (in concurrent drafting; this dispatch).



operational consequence: after-the-fact review of a hunt can answer “what authority did what, when, against what hypothesis” without ambiguity.

5. RUNBOOK POLICY BLOCK (PROPOSED)

The following SOAR-compatible runbook policy block operationalizes the four checkpoints. The block is illustrative; production deployments adapt the language to their SOAR platform’s policy vocabulary.

runbook_policy:

```

    name: "skipjack-hunt-cycle-reconciliation"
    version: "0.1-seed"
    authority: "soc-governance"
    applies_to: ["hunt-cycle-cases"]

    checkpoints:
      - id: "CP-1-triage-gate"
        phase_transition: "hunt → triage"
        trigger: "hunt_execution_complete"
        analyst_action_required: "classify_candidates_against_hypothesis"
        tool_recommendation: "permitted"
        auto_advance: "denied"
        audit_record: "classification + hypothesis_alignment_note"

      - id: "CP-2-escalation-gate"
        phase_transition: "triage → escalation"
        trigger: "candidate_classified_true_positive"
        analyst_action_required: "escalation_decision + consequence_review"
        threshold_consult: "ir_engagement_policy"
        auto_advance: "denied"
        audit_record: "escalation_decision + threshold_state + evidence_quality"

      - id: "CP-3-session-resumption-gate"
        phase_transition: "any → resume"
        trigger: "session_resume_event"
        analyst_action_required: "rehypothesize + state_review"
        auto_advance: "denied"
        audit_record: "resumption_event + analyst_authority_assertion"

      - id: "CP-4-scope-authority-gate"
        phase_transition: "scope_change OR stakeholder_change"
        trigger: "scope_event OR authority_event"
        analyst_action_required: "rehypothesize + scope_realignment"
        auto_advance: "denied"
        audit_record: "scope_event + new_authority_signature"

    governance:
      audit_retention: "P7Y"
      threshold_versioning: "required"

    policy_changes: "approval_chain_required"
```



The block is consumed by the SOAR platform as policy; agentic tooling operating within the hunt cycle respects the checkpoint structure; analyst decisions at each checkpoint are recorded against the policy version active.

6. LIMITATIONS AND CONCLUSION

6.1 LIMITATIONS

The four-checkpoint model assumes the hunt cycle is the dominant operational pattern. SOC's that operate primarily on alert-driven workflows (rather than hypothesis-driven hunts) may require adaptation; the alert-driven analog is left for future work.

The policy block in §5 is illustrative. Production SOAR platforms use varied policy languages and integration patterns; producing platform-specific variants is an engineering task this paper does not undertake.

Analyst capacity constraints may push back against checkpoint enforcement, particularly in volume-overwhelmed SOC's. The doctrine treats the constraint as a governance issue (under-resourced SOC) rather than as a reason to weaken the discipline.

6.2 CONCLUSION

The Skipjack reconciliation principle and the SOC hunt-cycle are isomorphic. The isomorphism yields four checkpoints (CP-1 through CP-4) that translate Skipjack discipline into SOC operational vocabulary. The result is an AI-assisted SOC doctrine that preserves analyst authority over the hypothesis, the triage decision, and the escalation call — at exactly the phase boundaries where automation bias would otherwise erode analyst authority.

This is Thesis Eight (cyb-08) of the HGC³AE² Cyber Pillar derivative family.

REFERENCES

— end v0.1-seed draft —

Drafting state: v0.1 CYB-08 (chain v0.1 step 5). 6-section canon shape with operational §5 runbook policy block. PR count: 12 — parent + automation-bias + Cummings + Sqrrl/Bianco hunt + ATT&CK + OWASP + SOAR/Chuvakin + NIST IR + Cyber Kill Chain + 3 cross-thesis refs (cyb-01, cyb-06, cyb-10). Meets ≥12 floor.